Porting GLASS
Python Array

ARCHER2 Celebration of Scienc

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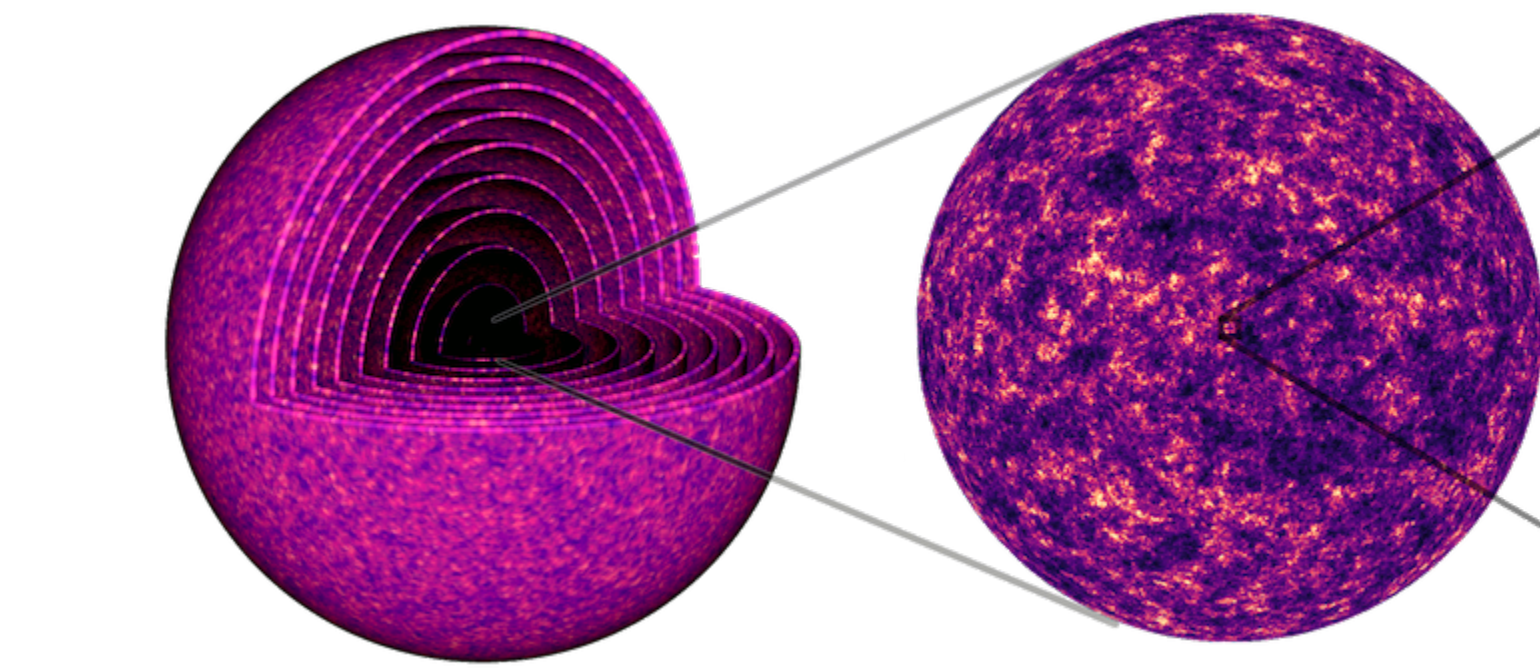
GLASS: Generato
Large Scale Struc

Overview

GLASS is a code used in cosmology to ge
of the full observable universe. In its curren
user base of the code are collaborations fo
surveys, which are a cornerstone of moder



The Team

Nicolas Tessore
(Principal Investigator + Technical Staff)

Connor Aird
(Technical Staff)

Saransh Chopra
(Technical Staff)

The Team

- Alessio Spurio Mancini (Co-Investigator)
- Arthur Loureiro (Co-Investigator)
- Benjamin Joachimi (Co-Investigator)
- Jason McEwen (Co-Investigator)
- Niall Jeffrey (Co-Investigator)
- Rebecca Martin (Research Administration

The Aim

1. Transform GLASS into a GPU-enabled coc
2. Improve the performance of simulations acceleration.
3. Implement and adapt parallel processing utilise GPU capabilities.
4. Optimise GLASS for modern GPU architec
5. Enable differentiable simulations using J full).
6. Embed GLASS within N-body simulations infrastructure for post-processing.

How Can This be Achieved

- Adopt the Python Array API Standard t
agnostic code.
- Core physics logic maps perfectly to the s
- The remaining gaps:
  - Handling RNG seeds consistently acros
backends.
  - Falling back to library-specific code for
  - Minimising the cost of moving arrays b
devices.

Python Array API Standard

What Are Arrays

- N-dimensional, grid-like data structure.
- `numpy.ndarray` is the most well known.
- “Rectangular” shape data type.
- Fast, easy to manipulate.
- Ubiquitous.

The Array Ecosystem

Aaron Meurer: Python Array API Standard, SciPy 2023

Motivation: End Users

- Ability to switch array libraries without re
code base.
- Avoid repeated transfers between array li
- Enable experimentation:
  - Test new hardware.
  - Use functionality specific to an array li

Motivation: Array *Providing*

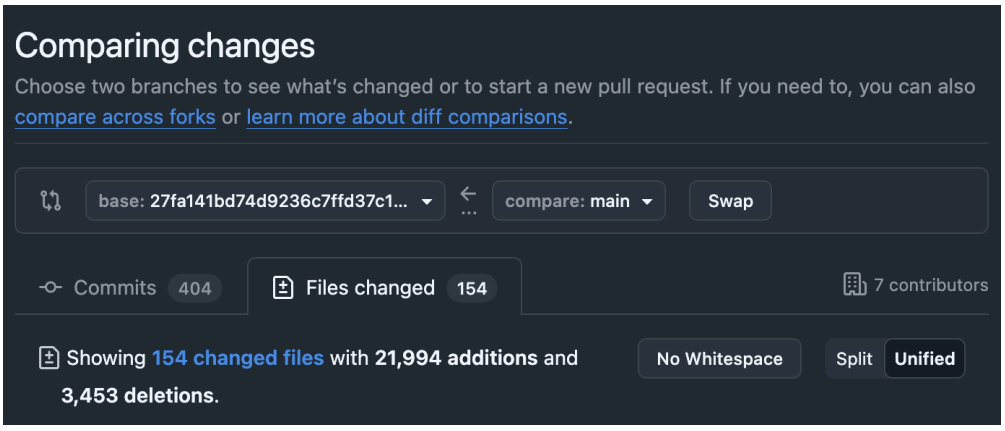
- Existing libraries:
  - Interoperability with new consuming li
  - Collaborative API decisions with other
- New libraries:
  - Concrete API to implement.
  - Guaranteed compatibility with consum

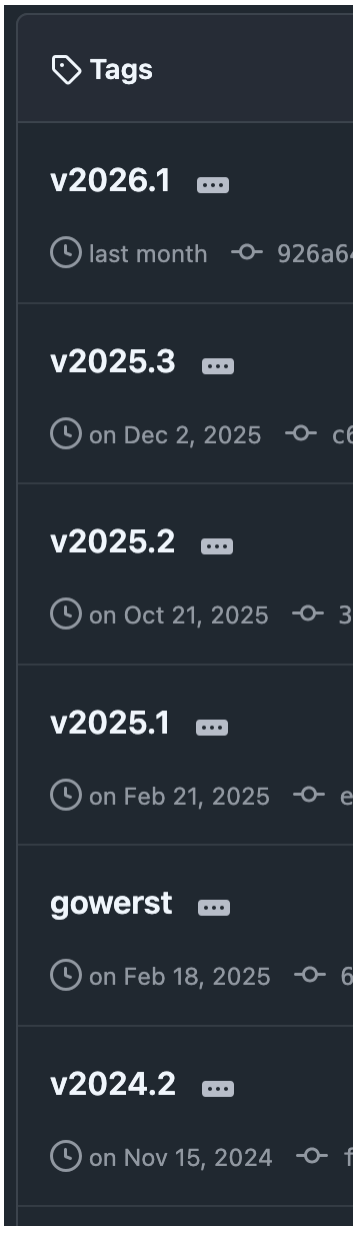
Motivation: Array *Consumi*

- Support hardware-accelerated and vend
  types.
- But without additional maintenance burd
- Library remains relevant when the comm
  new array library.

Progress

Changelog


Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff comparisons.
base: 27fa141bd74d9236c7ffd37c1...
compare: main
Swap
Commits 404
Files changed 154
7 contributors
Showing 154 changed files with 21,994 additions and 3,453 deletions.
No Whitespace
Split
Unified


Tags
v2026.1
last month
v2025.3
on Dec 2, 2025
v2025.2
on Oct 21, 2025
v2025.1
on Feb 21, 2025
gowerst
on Feb 18, 2025
v2024.2
on Nov 15, 2024

paddyroddy
260 commits 32,872 ++ 17,357 --
#1
Saransh-cpp
111 commits 10,101 ++ 3,787 --
#2
connoraird
64 commits 7,414 ++ 2,965 --
#3
pre-commit-ci[bot]
48 commits 4,007 ++ 1,286 --
#4
ntessore
35 commits 9,188 ++ 1,920 --
#5
dependabot[bot]
15 commits 33 ++ 33 --
#6
allcontributors[bot]
9 commits 139 ++ 20 --
#7
mwiet
1 commit 107 ++ 0 --
#8
Contributions
30
20
10
0
Jul '24
Jan '25
Jul '25
Jan '26

Testing on Different Array

./noxfile.py run as `uv run nox -s test`

```python
"""Nox config."""

import os
import pathlib
import shutil

import nox
import nox_uv

# Options to modify nox behaviour
nox.options.default_venv_backend = "uv"
nox.options.reuse_existing_virtualenvs = True
nox.options.sessions = [
    "lint",
    "tests",
]

ALL_PYTHON = [
```

Custom Typing

./glass/_types.py pending array-api-typin

```
1  from typing import TYPE_CHECKING, Any
2
3  if TYPE_CHECKING:
4      from collections.abc import Sequence
5      from typing import ParamSpec, TypeAlias, TypeV
6
7      import jaxtyping
8      import numpy as np
9
10     from array_api_strict._array_object import Arr
11     from array_api_strict._dtypes import DType
12
13     import glass.jax
14     from glass import _rng
15
16     P = ParamSpec("P")
17     R = TypeVar("R")
18     T = TypeVar("T")
```

Benchmarking

Perform regression tests by using pytest-be

```
Run regression test
199 ----------------------------------------------------------------------------
200 Performance has regressed:
201     test_generate[numpy-None] (0001_c4ad4a5) - Field 'mean' has failed PercentageRegressionChec
202     test_uniform_positions[numpy] (0001_c4ad4a5) - Field 'mean' has failed PercentageRegression
203 ----------------------------------------------------------------------------
204
205
206
207 ------------------------------------------------------ benchmark: 54 tests ---------------------
208 Name (time in us)                                                                        Mean
209 ---------------------------------------------------------------------------------------------------
210 test_cls2cov[numpy] (0001_c4ad4a5)                                                   402.0254 (2.85)
211 test_cls2cov[numpy] (NOW)                                                            344.6912 (2.44)
212 test_displacement[numpy] (0001_c4ad4a5)                                          103,969.9087 (737.3
213 test_displacement[numpy] (NOW)                                                   103,081.6571 (731.0
```

JAX does not allow array mutation, so need
regress NumPy as a result of enabling JAX.

Additional Array API Utiliti

glass/_array_api_utils.py for functions no
specification nor in array-api-extra .

```
"""
Array API Utilities for glass.
==============================

This module provides utility functions and classes
backends in the glass project, including NumPy, JA
functions for importing backends, determining arra
number generators, and providing missing functiona
xp_additions class.

Classes and functions in this module help ensure c
across different array libraries, and provide wrap
integration, interpolation, and linear algebra.

"""

from __future__ import annotations

```

Random Number Generato

glass/_rng.py

```
"""
Random Number Generation Utilities for glass.
=============================================

This module includes functions for dispatching ran
seeds. The chhoice of rng generator is determined
the user.

"""

from __future__ import annotations

from typing import TYPE_CHECKING

if TYPE_CHECKING:
    from types import ModuleType

    from glass.types import DTypeLike, FloatArray
```

JAX Support

./glass/jax.py

```python
"""Wrapper for JAX RNG with a NumPy-like interface

from __future__ import annotations

import math
import threading
from typing import TYPE_CHECKING

import jax.dtypes
import jax.numpy as jnp
import jax.random
import jax.scipy
import jax.typing

if TYPE_CHECKING:
    from jaxtyping import PRNGKeyArray
    from typing_extensions import Self

```

Further Work

- Test on CuPy following v14 release .
- Create scripts to benchmark on ARCHER2
array backends.
- Contribute remaining required functions
extra .
- Implement spherical harmonic transform
compatible.
- Tech debt…

https://
paddyr
talks